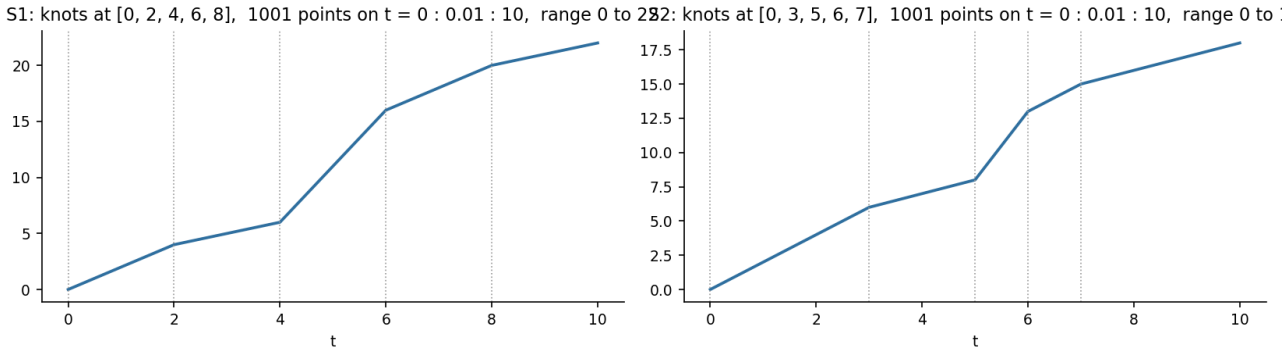
KAN approximation of S1 and S2: complete results

Experiments A (single edge, linear splines), B (single edge, cubic splines) and C (one hidden layer). Every fitted configuration is shown with its full curve, its error at every point, and its summary numbers. Data and generator: ExampleForORF.m.

Setup

Item	Value
Targets	S1: knots [0,2,4,6,8]; S2: knots [0,3,5,6,7]; slopes [2,-1,4,-3,-1] for both; S(t) = sum_j A_j * max(0, t - theta_j)
Sample	1001 points, t = 0, 0.01, ..., 10 (files S1.txt, S2.txt; regenerated from the script, match to 4e-15)
KAN library	PyKAN 0.2.8 (PyTorch 2.2.2), grid_range [0, 10], grid updating off, seed 0
Training	L-BFGS, 200 steps, full batch, target scaled to [0,1] for optimisation and rescaled after
Errors	computed on all 1001 points in original units; worst error = max |KAN - S|; RMSE = root mean square; mean abs = mean |KAN - S
Reproduce	python dump_fits.py (writes all_fits.json); python make_result_figs.py; python build_results_pdf.py



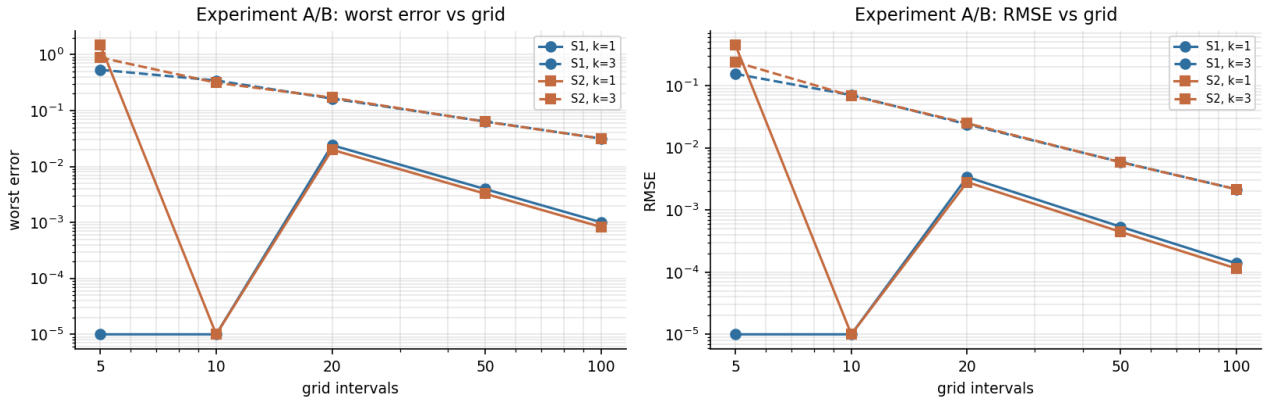
The two target functions. Dotted lines mark the knots.

1. All results in one table

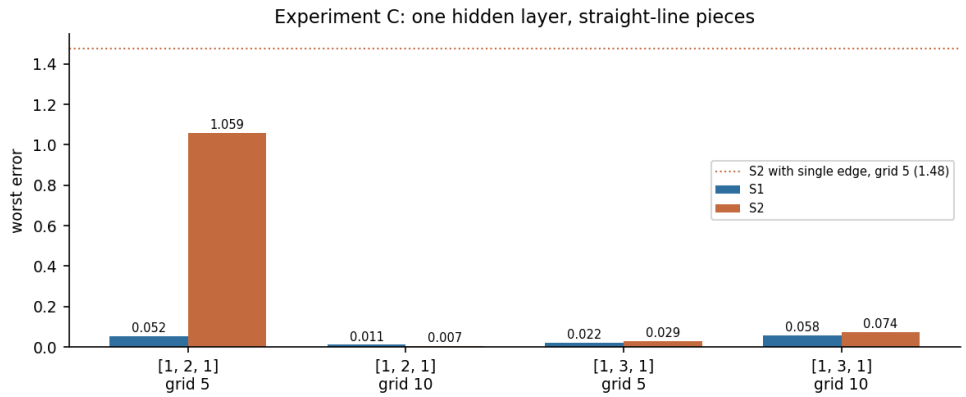
Sorted by experiment, then grid. 'Worst at t' is where the largest error occurs; for S2 on coarse grids it sits on a knot the grid cannot reach.

Exp	KAN width	Grid	Spline	Params	Fn	Worst error	RMSE	Mean abs	Worst at t
A	[1, 1]	5	linear k=1	12	S1	0 (exact)	0 (exact)	0 (exact)	7.27
A	[1, 1]	5	linear k=1	12	S2	1.4761	0.4475	0.3354	5.00
A	[1, 1]	10	linear k=1	17	S1	0 (exact)	0 (exact)	0 (exact)	6.22
A	[1, 1]	10	linear k=1	17	S2	0 (exact)	0 (exact)	0 (exact)	8.81
A	[1, 1]	20	linear k=1	27	S1	0.0240	0.0034	0.0014	0.00
A	[1, 1]	20	linear k=1	27	S2	0.0199	0.0028	0.0011	0.00
A	[1, 1]	50	linear k=1	57	S1	0.0039	0.0005	0.0002	0.20
A	[1, 1]	50	linear k=1	57	S2	0.0033	0.0004	0.0002	0.20
A	[1, 1]	100	linear k=1	107	S1	0.0010	0.0001	0 (exact)	0.10
A	[1, 1]	100	linear k=1	107	S2	0.0008	0.0001	0 (exact)	0.10
B	[1, 1]	5	cubic k=3	14	S1	0.5349	0.1542	0.1218	4.00
B	[1, 1]	5	cubic k=3	14	S2	0.8845	0.2394	0.1694	5.00
B	[1, 1]	10	cubic k=3	19	S1	0.3442	0.0703	0.0452	4.00
B	[1, 1]	10	cubic k=3	19	S2	0.3144	0.0684	0.0439	5.00
B	[1, 1]	20	cubic k=3	29	S1	0.1634	0.0238	0.0145	4.00
B	[1, 1]	20	cubic k=3	29	S2	0.1699	0.0248	0.0127	5.00
B	[1, 1]	50	cubic k=3	59	S1	0.0637	0.0059	0.0027	4.00
B	[1, 1]	50	cubic k=3	59	S2	0.0631	0.0059	0.0025	5.00
B	[1, 1]	100	cubic k=3	109	S1	0.0316	0.0022	0.0007	4.00
B	[1, 1]	100	cubic k=3	109	S2	0.0316	0.0022	0.0007	5.00
C	[1, 2, 1]	5	linear k=1	48	S1	0.0519	0.0183	0.0144	2.00
C	[1, 2, 1]	5	linear k=1	48	S2	1.0590	0.2815	0.1713	5.00
C	[1, 2, 1]	10	linear k=1	68	S1	0.0109	0.0019	0.0012	2.00
C	[1, 2, 1]	10	linear k=1	68	S2	0.0067	0.0018	0.0013	5.38
C	[1, 3, 1]	5	linear k=1	72	S1	0.0220	0.0034	0.0023	0.32
C	[1, 3, 1]	5	linear k=1	72	S2	0.0294	0.0052	0.0033	0.00
C	[1, 3, 1]	10	linear k=1	102	S1	0.0584	0.0128	0.0086	5.00
C	[1, 3, 1]	10	linear k=1	102	S2	0.0740	0.0168	0.0108	5.00

Summary charts



Experiments A and B: worst error (left) and RMSE (right) against grid size, log scales. Solid = linear splines, dashed = cubic.



Experiment C: worst error by configuration. Dotted line: the single-edge grid-5 result on S2, for comparison.

2. S1 versus S2, side by side

Experiment A: single edge [1,1], linear splines (k=1)

KAN width	Grid	Params	S1 worst	S1 RMSE	S2 worst	S2 RMSE	S2 / S1 worst
[1, 1]	5	12	0 (exact)	0 (exact)	1.4761	0.4475	n/a
[1, 1]	10	17	0 (exact)	0 (exact)	0 (exact)	0 (exact)	n/a
[1, 1]	20	27	0.0240	0.0034	0.0199	0.0028	0.8x
[1, 1]	50	57	0.0039	0.0005	0.0033	0.0004	0.8x
[1, 1]	100	107	0.0010	0.0001	0.0008	0.0001	0.8x

Experiment B: single edge [1,1], cubic splines (k=3)

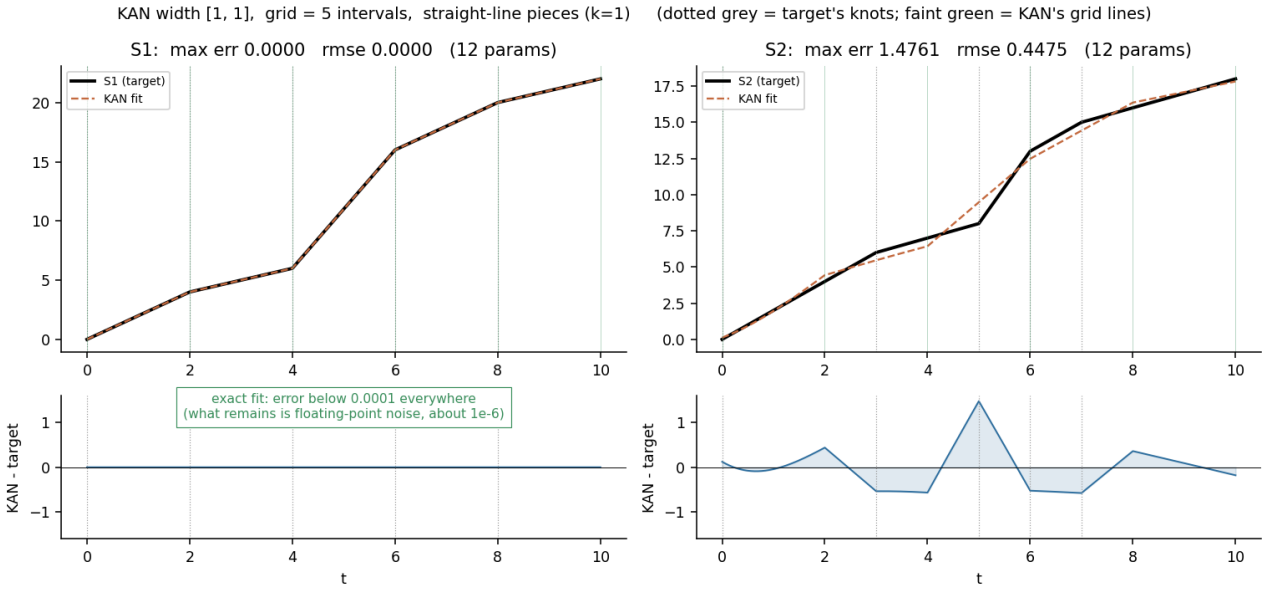
KAN width	Grid	Params	S1 worst	S1 RMSE	S2 worst	S2 RMSE	S2 / S1 worst
[1, 1]	5	14	0.5349	0.1542	0.8845	0.2394	1.7x
[1, 1]	10	19	0.3442	0.0703	0.3144	0.0684	0.9x
[1, 1]	20	29	0.1634	0.0238	0.1699	0.0248	1.0x
[1, 1]	50	59	0.0637	0.0059	0.0631	0.0059	1.0x
[1, 1]	100	109	0.0316	0.0022	0.0316	0.0022	1.0x

Experiment C: one hidden layer, linear splines (k=1)

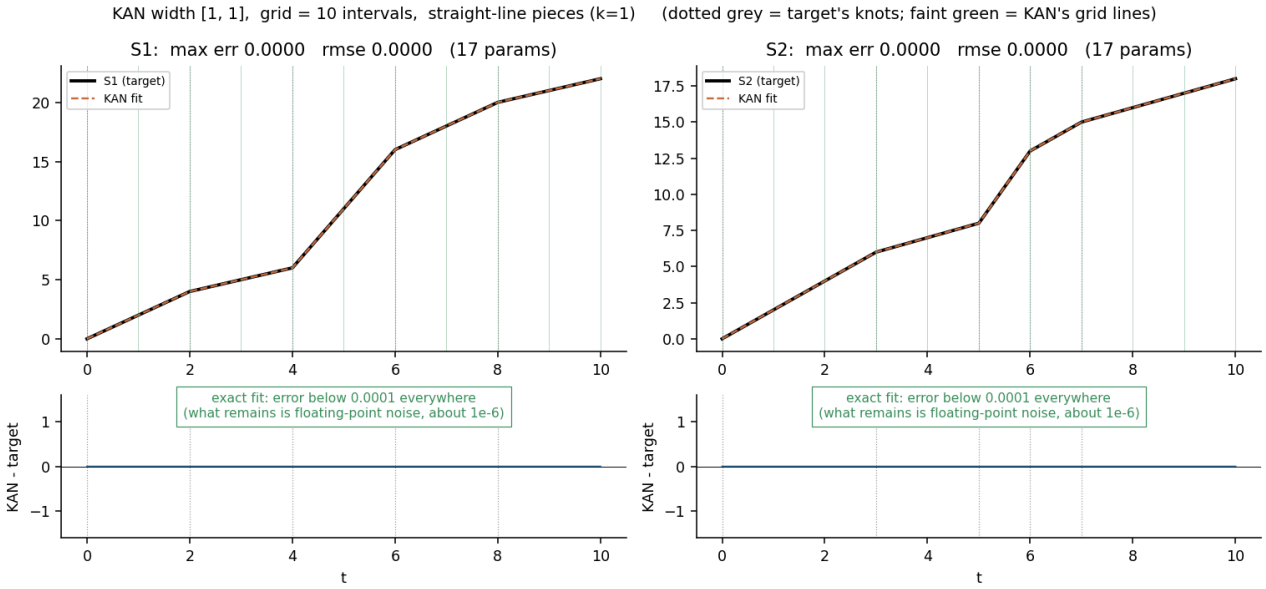
KAN width	Grid	Params	S1 worst	S1 RMSE	S2 worst	S2 RMSE	S2 / S1 worst
[1, 2, 1]	5	48	0.0519	0.0183	1.0590	0.2815	20.4x
[1, 2, 1]	10	68	0.0109	0.0019	0.0067	0.0018	0.6x
[1, 3, 1]	5	72	0.0220	0.0034	0.0294	0.0052	1.3x
[1, 3, 1]	10	102	0.0584	0.0128	0.0740	0.0168	1.3x

3. Every fit, one configuration per figure

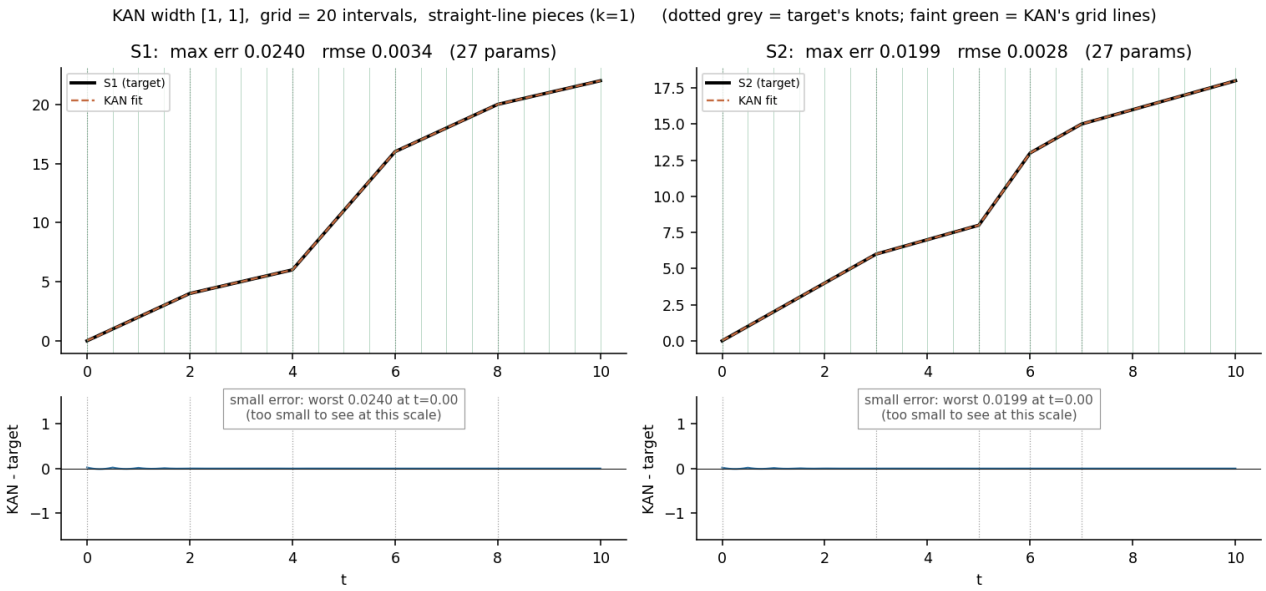
Top row: target (black) and KAN fit (orange dashed), with the target's knots (dotted grey) and the KAN's grid lines (faint green). Bottom row: the error KAN - target at every point. Every error panel uses the same vertical scale, -1.6 to +1.6, so figures can be compared directly: a flat line at zero is a perfect fit, a wave is a real error. Where the error is too small to show at this scale, the panel says so and gives the number.



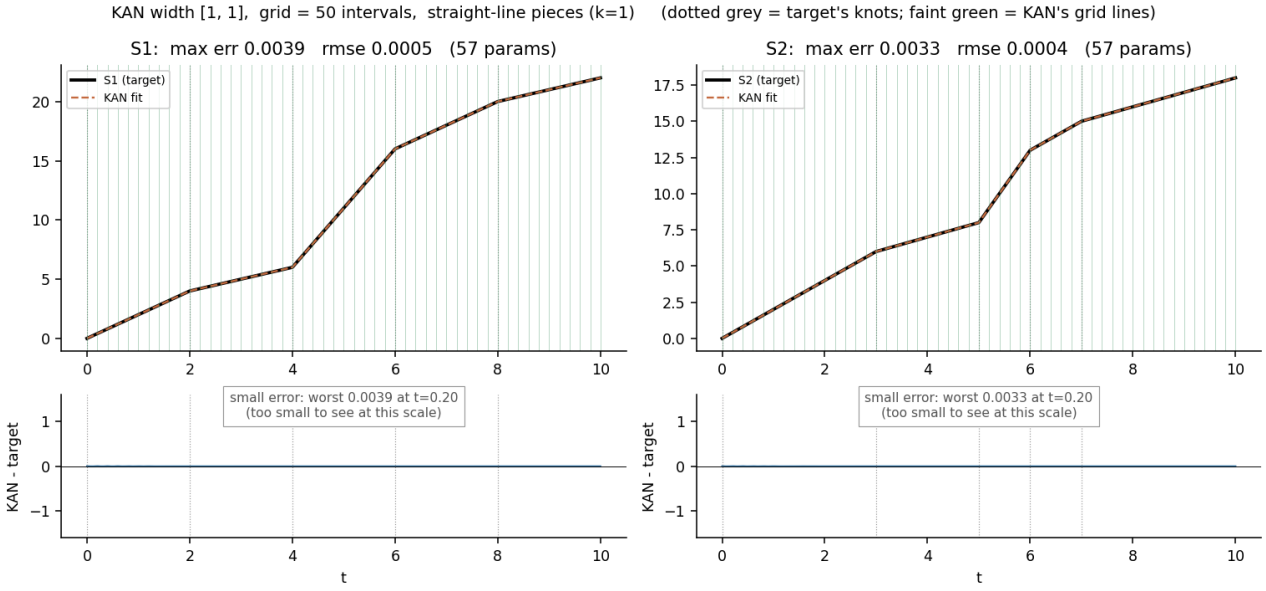
Experiment A. Width [1, 1], grid 5, k=1, 12 parameters. S1: worst 0 (exact), RMSE 0 (exact), worst at t=7.27. S2: worst 1.4761, RMSE 0.4475, worst at t=5.00.



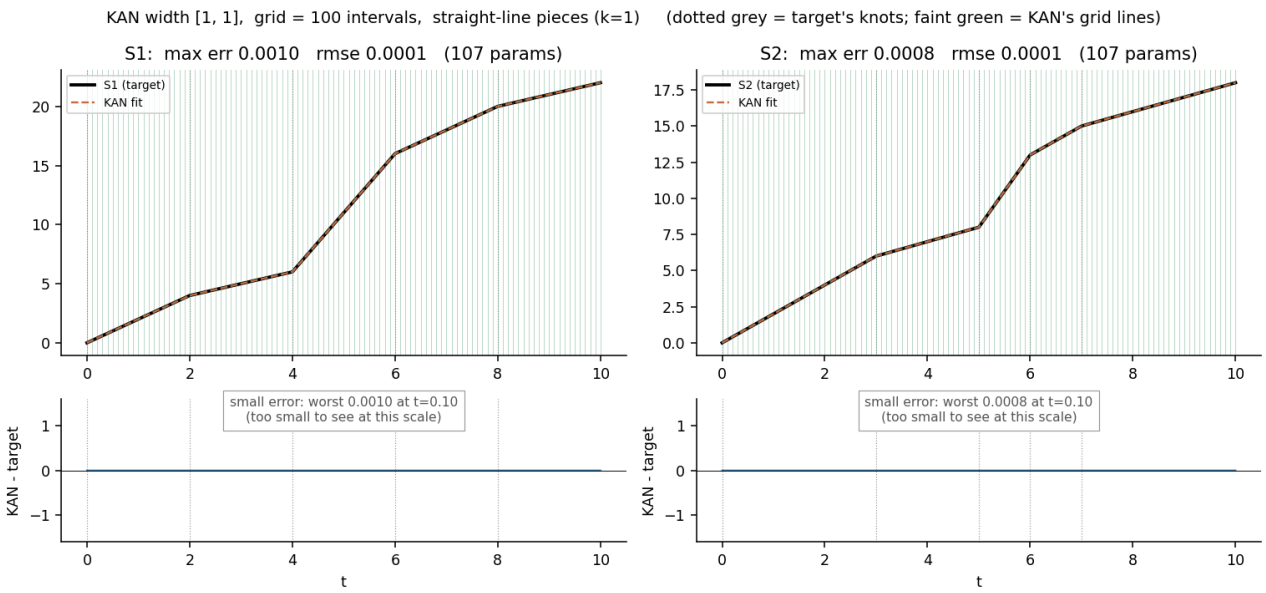
Experiment A. Width [1, 1], grid 10, k=1, 17 parameters. S1: worst 0 (exact), RMSE 0 (exact), worst at t=6.22. S2: worst 0 (exact), RMSE 0 (exact), worst at t=8.81.



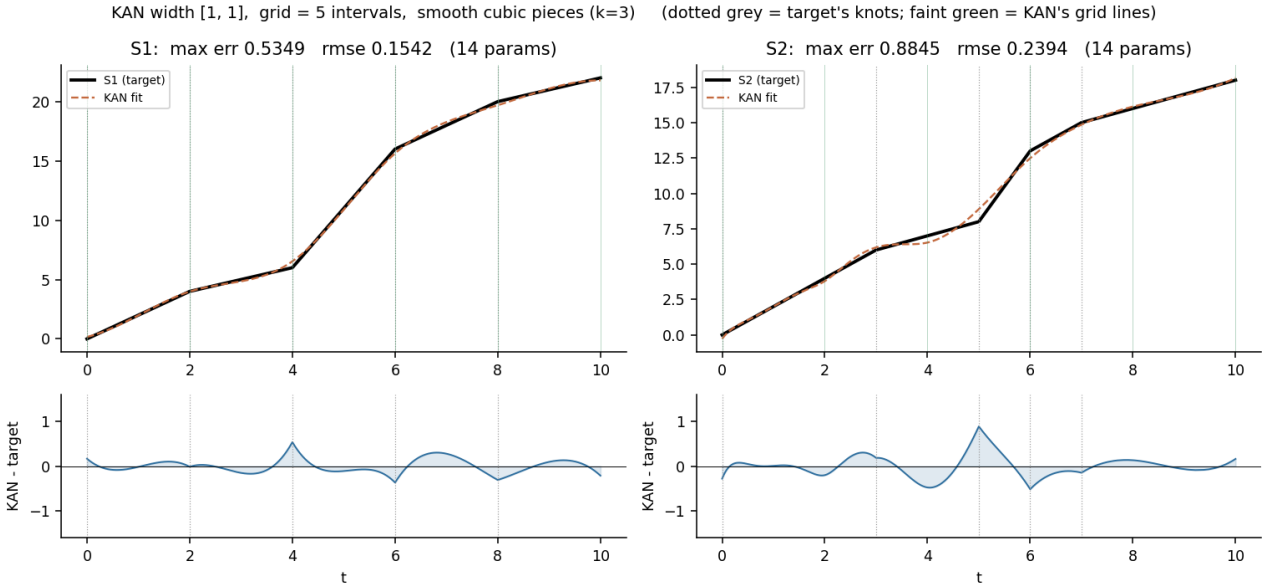
Experiment A. Width [1, 1], grid 20, k=1, 27 parameters. S1: worst 0.0240, RMSE 0.0034, worst at t=0.00. S2: worst 0.0199, RMSE 0.0028, worst at t=0.00.



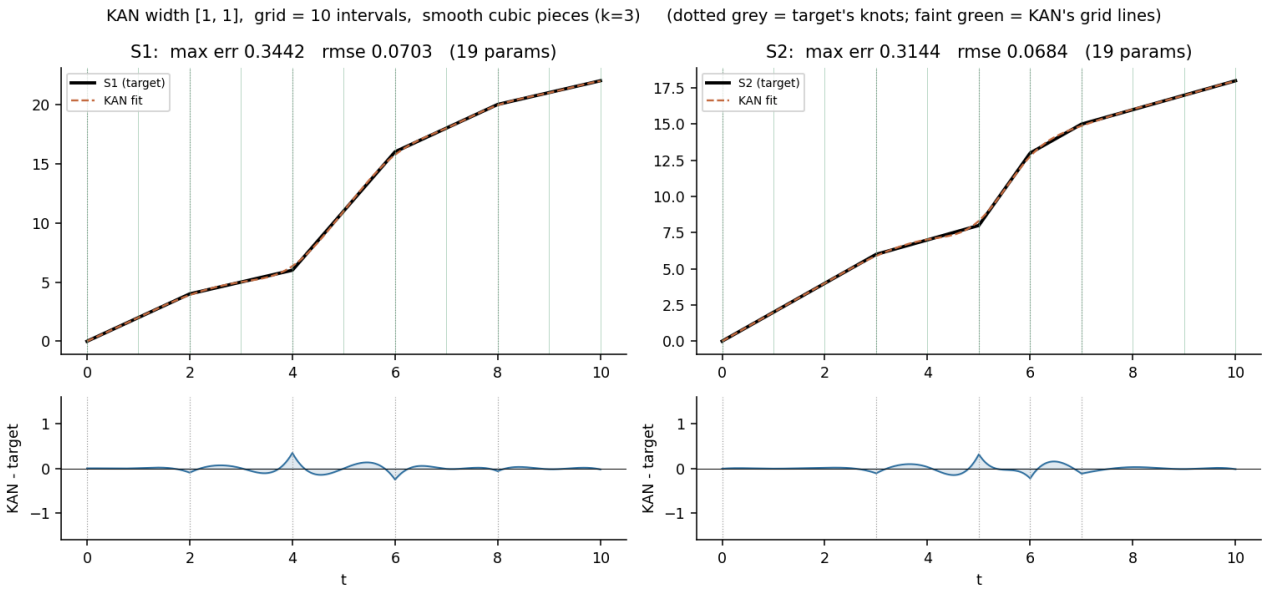
Experiment A. Width [1, 1], grid 50, k=1, 57 parameters. S1: worst 0.0039, RMSE 0.0005, worst at t=0.20. S2: worst 0.0033, RMSE 0.0004, worst at t=0.20.



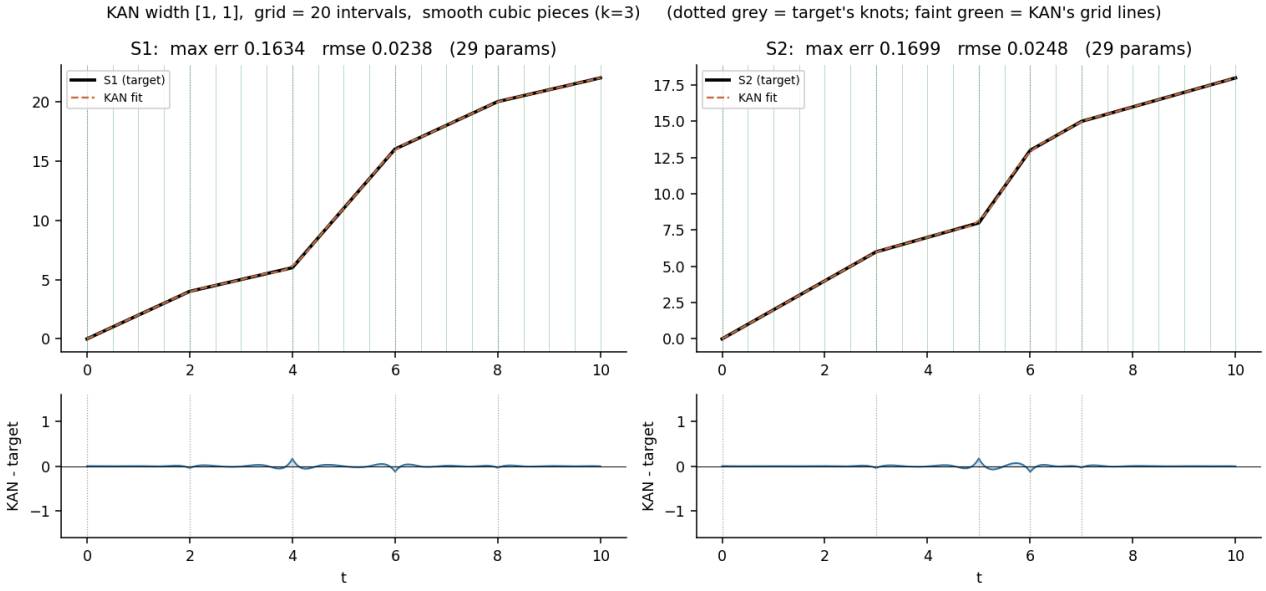
Experiment A. Width [1, 1], grid 100, k=1, 107 parameters. S1: worst 0.0010, RMSE 0.0001, worst at t=0.10. S2: worst 0.0008, RMSE 0.0001, worst at t=0.10.



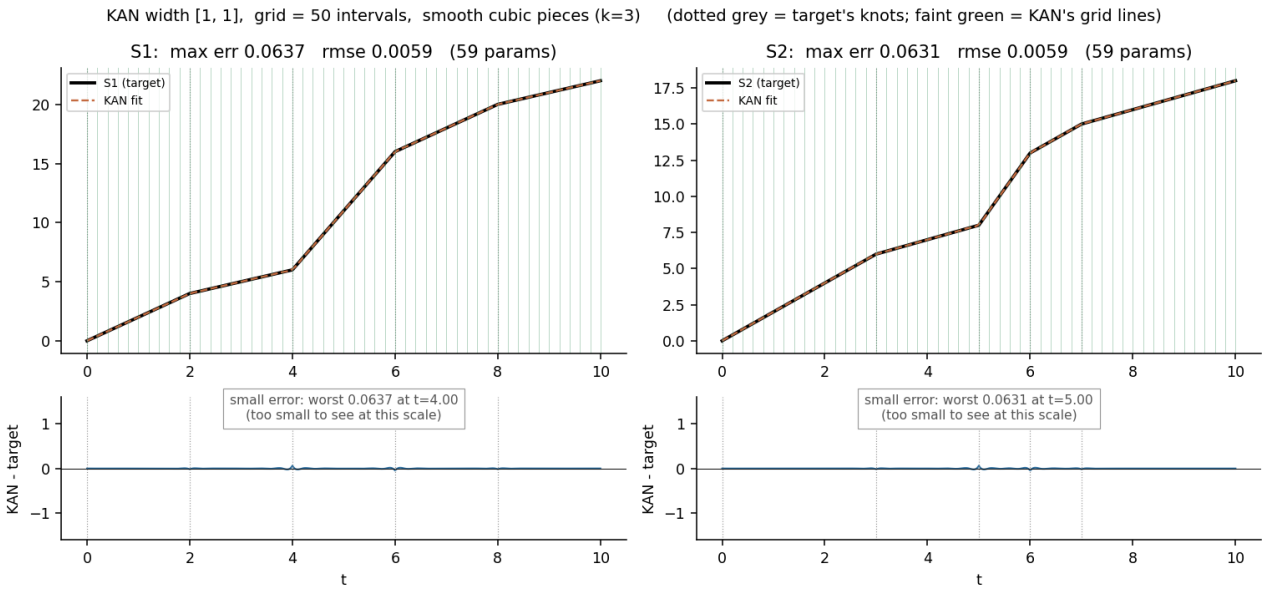
Experiment B. Width [1, 1], grid 5, k=3, 14 parameters. S1: worst 0.5349, RMSE 0.1542, worst at t=4.00. S2: worst 0.8845, RMSE 0.2394, worst at t=5.00.



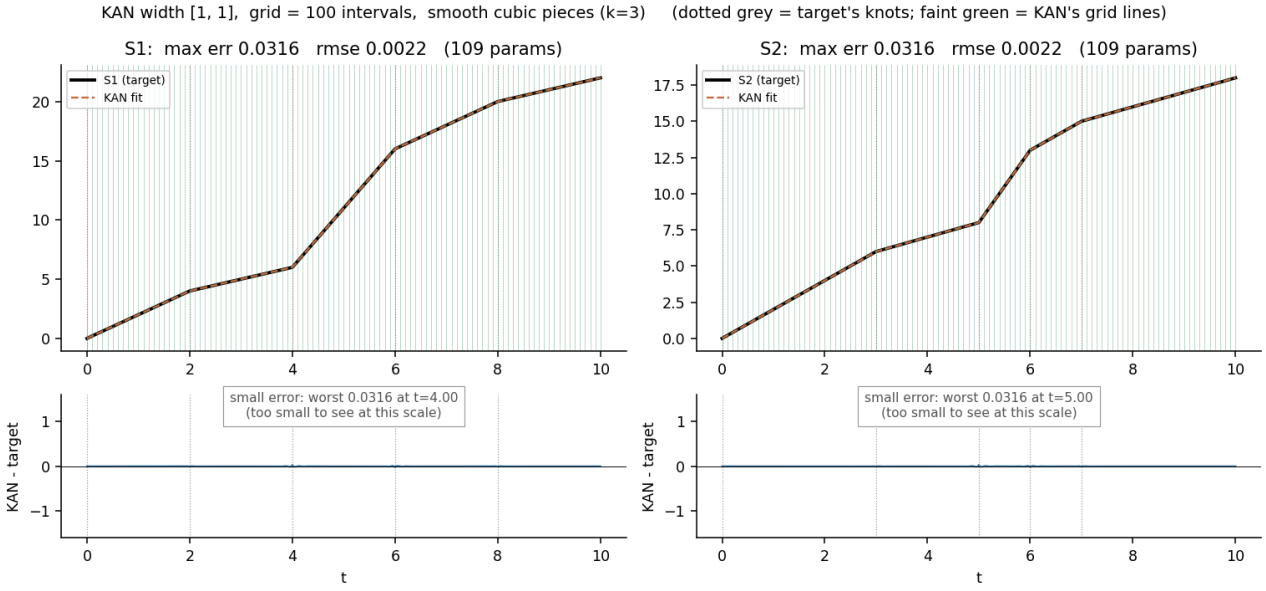
Experiment B. Width [1, 1], grid 10, k=3, 19 parameters. S1: worst 0.3442, RMSE 0.0703, worst at t=4.00. S2: worst 0.3144, RMSE 0.0684, worst at t=5.00.



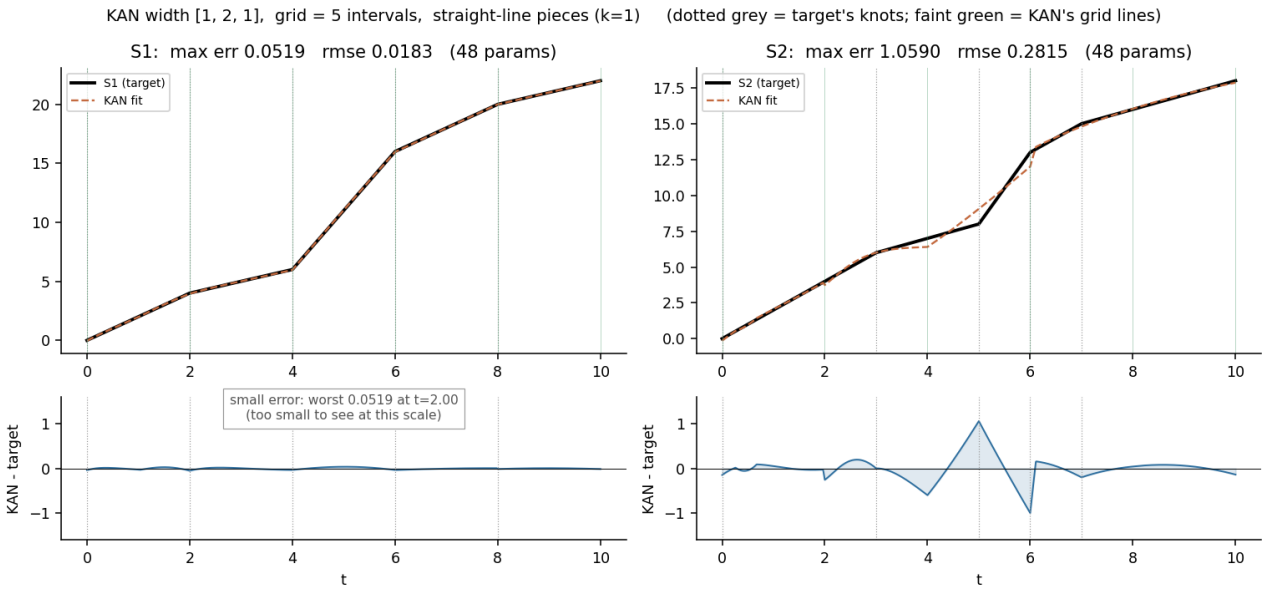
Experiment B. Width [1, 1], grid 20, k=3, 29 parameters. S1: worst 0.1634, RMSE 0.0238, worst at t=4.00. S2: worst 0.1699, RMSE 0.0248, worst at t=5.00.



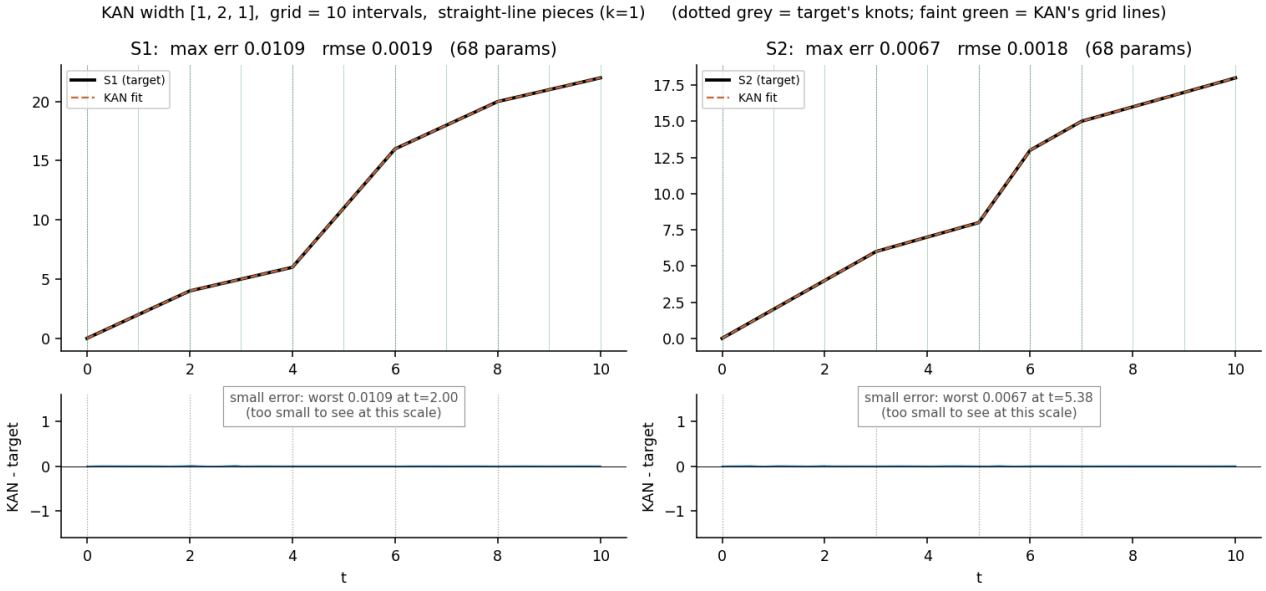
Experiment B. Width [1, 1], grid 50, k=3, 59 parameters. S1: worst 0.0637, RMSE 0.0059, worst at t=4.00. S2: worst 0.0631, RMSE 0.0059, worst at t=5.00.



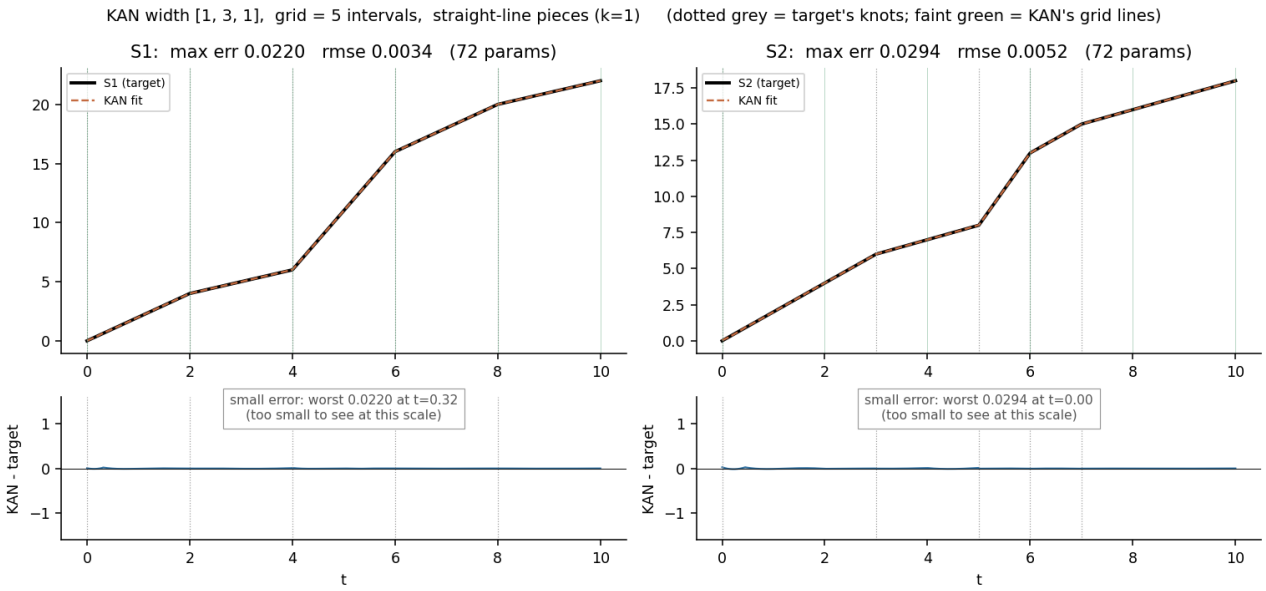
Experiment B. Width [1, 1], grid 100, k=3, 109 parameters. S1: worst 0.0316, RMSE 0.0022, worst at t=4.00. S2: worst 0.0316, RMSE 0.0022, worst at t=5.00.



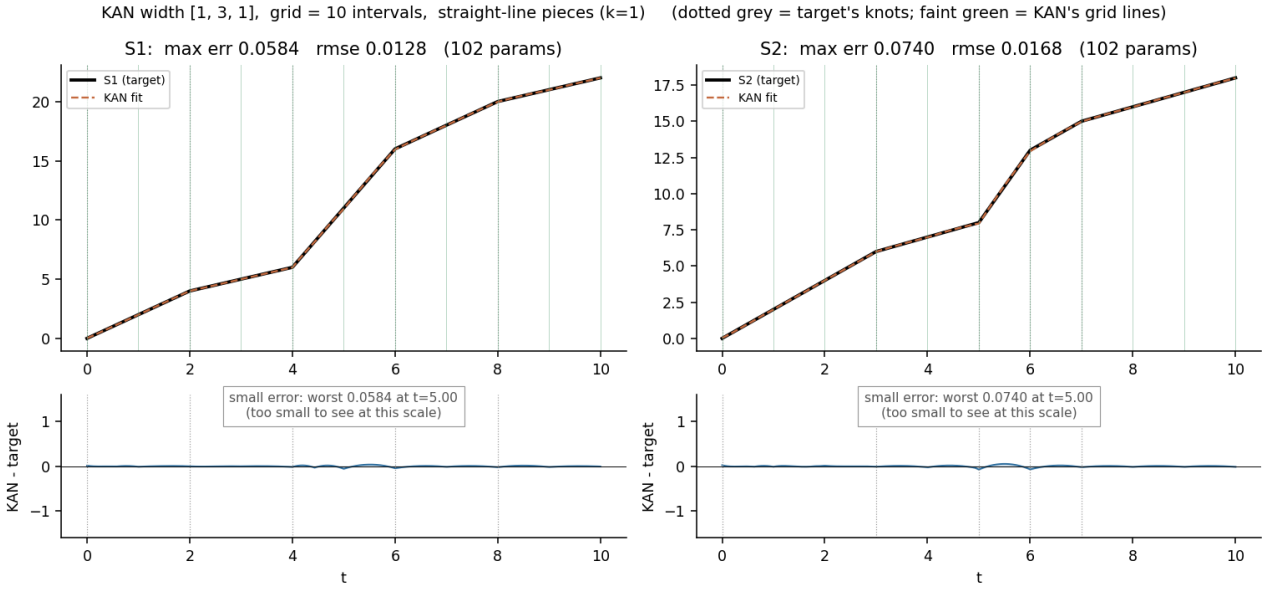
Experiment C. Width [1, 2, 1], grid 5, k=1, 48 parameters. S1: worst 0.0519, RMSE 0.0183, worst at t=2.00. S2: worst 1.0590, RMSE 0.2815, worst at t=5.00.



Experiment C. Width [1, 2, 1], grid 10, k=1, 68 parameters. S1: worst 0.0109, RMSE 0.0019, worst at t=2.00. S2: worst 0.0067, RMSE 0.0018, worst at t=5.38.



Experiment C. Width [1, 3, 1], grid 5, k=1, 72 parameters. S1: worst 0.0220, RMSE 0.0034, worst at t=0.32. S2: worst 0.0294, RMSE 0.0052, worst at t=0.00.



Experiment C. Width [1, 3, 1], grid 10, k=1, 102 parameters. S1: worst 0.0584, RMSE 0.0128, worst at t=5.00. S2: worst 0.0740, RMSE 0.0168, worst at t=5.00.